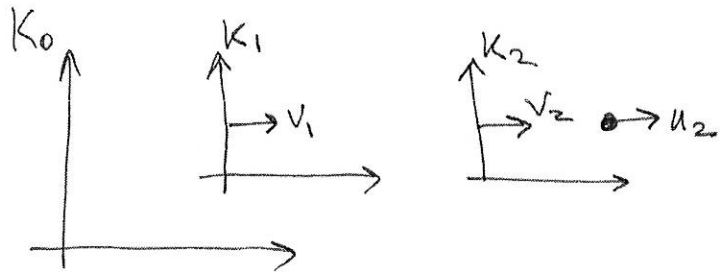
(P1)

a)



Object moves with u_2 along $\hat{x}$ in frame K_2.
In frame K_1 it moves with $u_1 = \dfrac{u_2+v_2}{1+\frac{u_2 v_2}{c^2}}$

In frame K_0: $u_0 = \dfrac{u_1+v_1}{1+\frac{u_1 v_1}{c^2}}$

$$u_0 = \frac{u_1+v_1}{1+\frac{u_1 v_1}{c^2}} = \frac{\frac{u_2+v_2}{1+\frac{u_2 v_2}{c^2}} + v_1}{1+\frac{v_1}{c^2}\left(\frac{u_2+v_2}{1+\frac{u_2 v_2}{c^2}}\right)} =$$

$$= \frac{\left[u_2+v_2+v_1\left(1+\frac{u_2 v_2}{c^2}\right)\right]\Big/\left(1+\frac{u_2 v_2}{c^2}\right)}{\cancel{\left[1+\frac{u_2+v_2}{1+\frac{u_2 v_2}{c^2}}+\frac{v_1}{c^2}\right]\left(1+\frac{u_2 v_2}{c^2}\right)}} =$$

$$\left[1+\frac{u_2 v_2}{c^2}+\frac{v_1}{c^2}(u_2+v_2)\right]\Big/\left(1+\frac{u_2 v_2}{c^2}\right)$$

$$= \frac{u_2+v_2+v_1+\frac{v_1 u_2 v_2}{c^2}}{1+\frac{v_1(u_2+v_2)+u_2 v_2}{c^2}} = \frac{u_2\left(1+\frac{v_1 v_2}{c^2}\right)+v_1+v_2}{1+\frac{v_1 v_2}{c^2}+\frac{u_2(v_1+v_2)}{c^2}}$$

$$u_0 = \frac{\left(1+\frac{v_1 v_2}{c^2}\right)\left(u_2 + \frac{v_1+v_2}{1+\frac{v_1 v_2}{c^2}}\right)}{\left(1+\frac{v_1 v_2}{c^2}\right)\left(1+\frac{u_2(v_1+v_2)}{c^2\left(1+\frac{v_1 v_2}{c^2}\right)}\right)} =$$

$$= \frac{u_2 + \tilde{v}}{1+\frac{u_2 \tilde{v}}{c^2}} \quad , \text{where } \tilde{v} = \frac{v_1+v_2}{1+\frac{v_1 v_2}{c^2}}$$

is the velocity of K_2 with respect to K_0.

$\tilde{v}$ is just relativistic formula of velocity addition.

(b) Let us introduce rapidity ψ as: $\tanh(\psi) = \frac{v}{c}$

Then $\tanh(\psi_1) = \frac{v_1}{c}$, $\tanh(\psi_2) = \frac{v_2}{c}$

Then $\frac{\tilde{v}}{c} = \frac{\frac{v_1+v_2}{c}}{1+\frac{v_1 v_2}{c^2}} \Rightarrow \tanh(\tilde{\psi}) = \frac{\tanh\psi_1 + \tanh\psi_2}{1+\tanh\psi_1 \tanh\psi_2} =$

$$= \tanh(\psi_1+\psi_2)$$

$\Rightarrow \tilde{\psi} = \psi_1 + \psi_2$! Rapidies just add

(c) With n frames, $\frac{u_0}{c} = \tanh(\tilde{\psi}) = \tanh(\psi_1 + \psi_2 + \dots \psi_n + \psi_{n+1})$

due to u_n

$$= \tanh((n+1)\psi) = \tanh\left[(n+1)\tanh^{-1}\left(\frac{v}{c}\right)\right] \xrightarrow[n\to\infty]{} 1$$

All rapidities are the same.

$$\tanh(\psi) = \frac{v}{c}$$

(P2) Earth

$t=0$: $A\ V_A \rightarrow$ $\quad$ $\leftarrow V_B\ B$

X_A $\quad$ X_B

$V_A = 0.8c$

$V_B = 0.6c$

$X_B - X_A = \ell = 4.2 \cdot 10^8 m$

a)

$$\left.\begin{aligned} X_A(t) &= X_A + V_A t \\ X_B(t) &= X_B - V_B t \end{aligned}\right\} \quad X_A + V_A t = X_B - V_B t$$

$$t = \frac{X_B - X_A}{V_A + V_B} = \frac{4.2 \cdot 10^8 m}{1.4c}$$

$$t = 1 s.$$

b) Transform to frame $K' = K_A$ of rocket A

$$x' = \frac{x - V_A t}{\sqrt{1 - V_A^2/c^2}}$$

$$t' = \frac{t - \frac{V_A}{c^2} x}{\sqrt{1 - V_A^2/c^2}}$$

Velocity of B in K' $\quad v'_B = \frac{dx'}{dt'} = \frac{dx - V_A dt}{dt - \frac{V_A}{c^2} dx} =$

$$= \frac{\frac{dx}{dt} - V_A}{1 - \frac{V_A}{c^2}\frac{dx}{dt}} = \frac{V_B - V_A}{1 - \frac{V_A V_B}{c^2}} = \frac{(-0.6 - 0.8)c}{1 - (0.8)(-0.6)} = \frac{-1.4c}{1.48}$$

$$V_B = \frac{dx}{dt} = -0.6c$$

$$v'_B \cong -0.95c$$

c) In $K' = K_A$, $x' = 0 \Rightarrow x = V_A t \Rightarrow t' = \frac{t - \frac{V_A}{c^2}(V_A t)}{\sqrt{1 - V_A^2/c^2}} =$

$$= \sqrt{1 - \frac{V_A^2}{c^2}}\, t = 0.6 s \text{ (proper time)}$$

Problem 4

Velocity transformations

$$v_x = \frac{v_x' + V}{1+\beta \frac{v_x'}{c}} \qquad v_y = \frac{v_y'}{\gamma\left(1+\beta \frac{v_x'}{c}\right)} \qquad \beta = \frac{V}{c}, \quad \gamma = \frac{1}{\sqrt{1-\beta^2}}$$

Velocity differentials

$$dv_x = \frac{dv_x'}{\gamma^2\left(1+\frac{\beta v_x'}{c}\right)^2}$$

$$dv_y = \frac{dv_y'}{\gamma\left(1+\beta \frac{v_x'}{c}\right)} - \frac{v_y' \frac{\beta}{c} dv_x'}{\gamma\left(1+\beta \frac{v_x'}{c}\right)^2}$$

Time differential

$$dt = \gamma dt'\left(1+\frac{\beta v_x'}{c}\right)$$

Acceleration ($a_x = \frac{dv_x}{dt}$, $a_x' = \frac{dv_x'}{dt'}$, etc.)

$$a_x = \frac{a_x'}{\gamma^3\left(1+\beta \frac{v_x'}{c}\right)^3}$$

$$a_y = \frac{a_y'}{\gamma^2\left(1+\beta \frac{v_x'}{c}\right)^2} - \frac{\frac{\beta}{c} v_y' a_x'}{\gamma^2\left(1+\beta \frac{v_x'}{c}\right)^3}$$